

量子力学期中考试 (取 $\hbar=1$)

一、

(1) (x, y, p_z) 在动量表象下共同

(4')

本征波函数

$$\psi_x = e^{-ip_x x_0} \quad \psi_y = e^{-ip_y y_0} \quad \psi_z = \delta(p_z - p_0)$$

x, y, p_z 互相对易

所以乘起来就是

(2)
$$V(x, y, z) = \begin{cases} 0 & z \leq 0 \\ V_0 & z > 0 \end{cases}$$

(4')

给出体系
守恒量

算符守恒量

$$\frac{dA}{dt} = \frac{\partial A}{\partial t} + \frac{1}{i\hbar} [A, H]$$

'' 为守恒

即要与 $\hat{H}$ 对易

$$\hat{H} = \frac{1}{2} (p_x^2 + p_y^2 + p_z^2) + V(z)$$

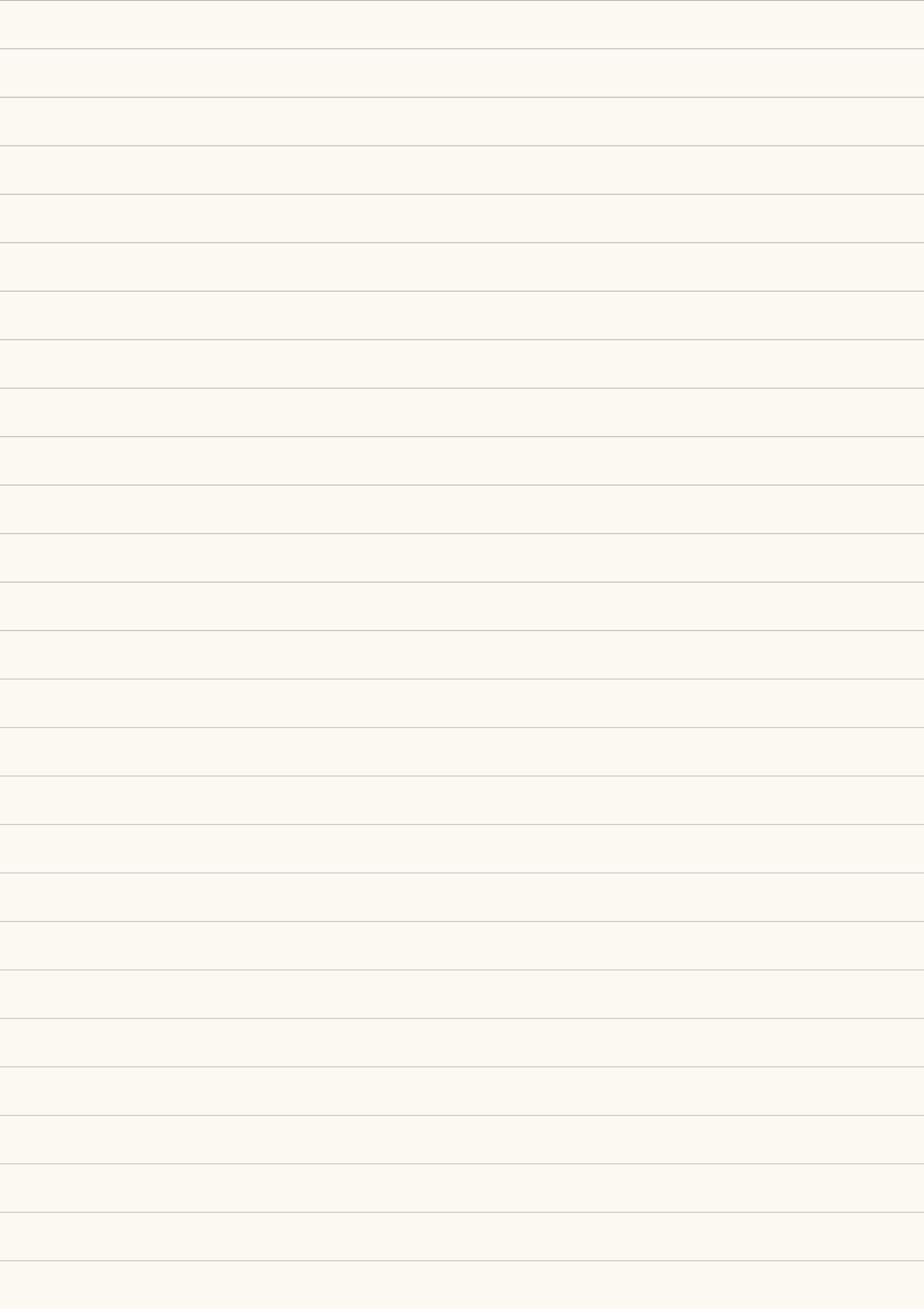
① H ② p_x ③ p_y ④ L_z

(3) $H = \frac{1}{2} p^2 - \chi$ 已知 $t=0$ $\bar{x} = x_0$ $\bar{p} = p_0$

(6') 求 $\bar{x}(t)$ $\bar{p}(t)$

$$\frac{d\bar{x}}{dt} = \frac{1}{i} \overline{[x, H]} = \frac{1}{i} \overline{[\hat{x}, \frac{p^2}{2}]} = \bar{p}$$

$$\frac{d\bar{p}}{dt} = \frac{1}{i} \overline{[\hat{p}, H]} = 0$$



$$\therefore \bar{p} = t + p_0$$

$$\bar{x} = \frac{1}{2}t^2 + p_0 t + x_0$$

(4) 同(3), 但是 $H(x) = \frac{1}{2}p^2 - p$
(16')

$$\frac{d\bar{x}}{dt} = p - 1$$

$$\therefore \bar{x} = x_0 + (p_0 - 1)t$$

$$\frac{dp}{dt} = 0$$

$$\bar{p} = p_0$$

二
(20') $V(x) = 4x^4$, 处于基态

(16'11) 求基态宇称, 理由

16'12) 求基态 $\bar{x}$ $\bar{p}$

18'13) 用 $\Delta x \cdot \Delta p_x = \frac{1}{2}$ 估算基态能量

(1) $[\hat{H}, \hat{p}] = 0$ $\therefore$ 基态具有确定宇称

(3') 理由: 基态无节点, 即除了无穷远无零点
所以偶宇称 (3')

$$(2) \langle x \rangle = \int_{-\infty}^{+\infty} \psi^* x \psi dx = 0$$

$$\langle p \rangle = \dots = 0$$

$$(3) \Delta x = \sqrt{\overline{x^2} - \bar{x}^2} = \sqrt{\overline{x^2}}$$

$$\therefore \overline{x^2} \cdot \overline{p^2} = \frac{1}{4}$$

$$H \approx \frac{1}{2} \bar{p}^2 + 4(\bar{x}^2)^2$$

$$= \frac{1}{2} \bar{p}^2 + 4 \left(\frac{1}{4} \frac{1}{\bar{p}^2} \right)^2$$

$$\frac{dH}{dp} = 0 \text{ 即可}$$

$$H = E = \frac{3}{4}$$

五. $V(x) = \frac{1}{2}x^2$, V_c 为本征态, U_n 为能量本征态

$$\hat{a} V_c = c V_c$$

(1) $\hat{O} = \hat{x} \hat{p}_x - \hat{p}_x \hat{x}$ 求 U_2 的 $\bar{O}$

$$\hat{x} = \frac{1}{\sqrt{2}} (a^\dagger + a) \quad \hat{p} = \frac{1}{\sqrt{2}i} (a - a^\dagger)$$

$$\begin{aligned} (4') \quad \hat{O} &= \frac{1}{2i} (a^\dagger a + a^2 - a^{\dagger 2} - a a^\dagger + a a^\dagger - a^{\dagger 2} \\ &\quad + a^2 - a^\dagger a) \\ &= \frac{1}{i} (a^2 - a^{\dagger 2}) \end{aligned}$$

$$\begin{aligned} \bar{O} &= \int U_2^* \hat{O} U_2 dx = \int U_2^* \frac{1}{i} (\sqrt{2} \times 1 U_0 - \sqrt{3} \times 4 U_4) dx \\ &= 0 \end{aligned}$$

(2) 同 (1), 求 V_c 的 $\bar{0}$

$$\bar{0} = -i \int V_c^* (a^2 - a^{\dagger 2}) V_c dx$$

$$= -i (c^2 - c^{*2})$$

(3) 求 $V_c = \bar{N}, \bar{N}^2, \Delta N$

$$\bar{N} = a^\dagger a$$

$$\bar{N} = \int V_c^* a^\dagger a V_c dx$$

$$= \int (a V_c)^* a V_c dx$$

$$= |c|^2$$

$$\bar{N}^2 = \int V_c^* a^\dagger a a^\dagger a V_c dx$$

$$= \int V_c^* c^* a a^\dagger c V_c dx$$

利用 $a a^\dagger = a^\dagger a + 1$

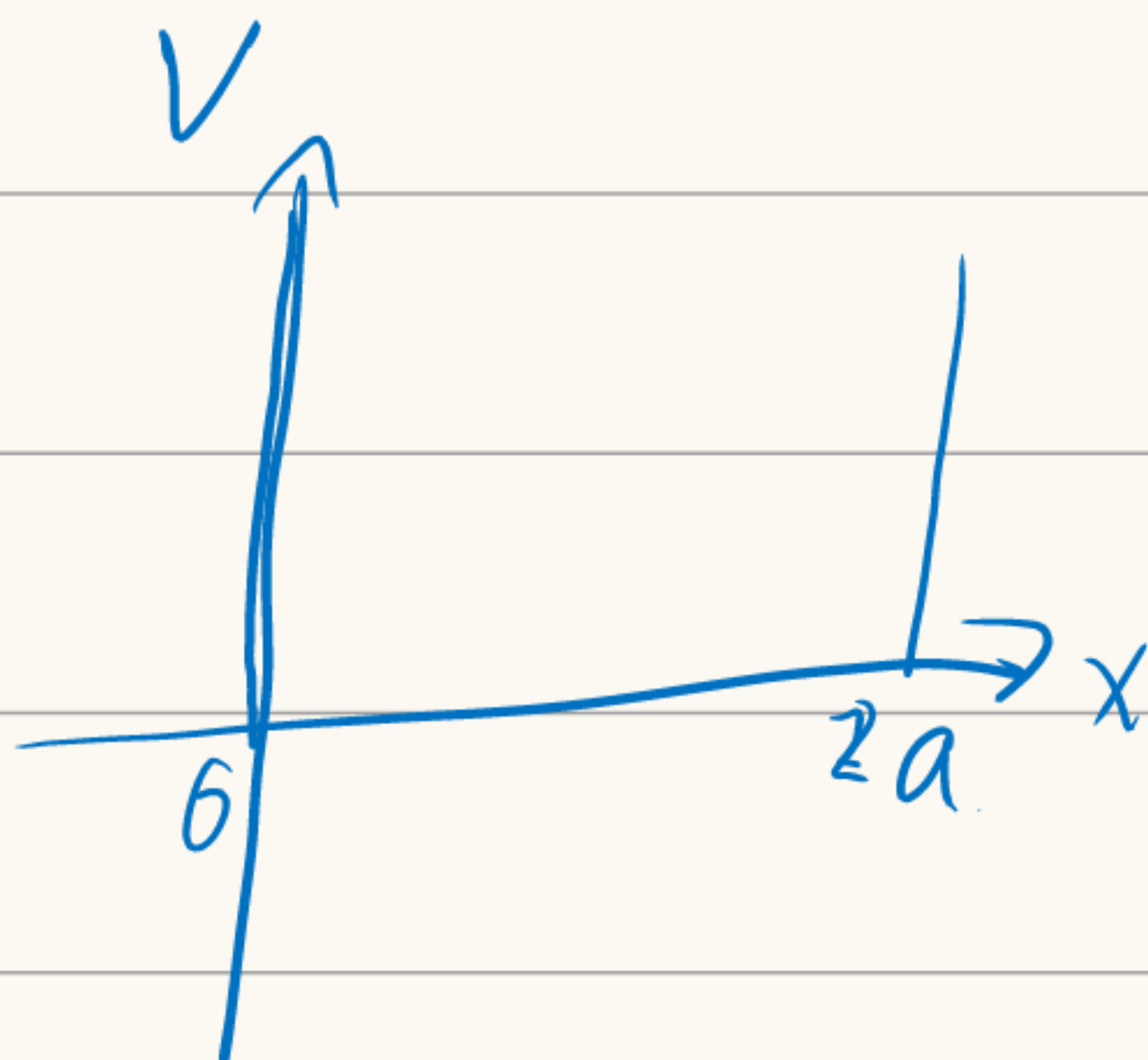
$$= \int |c|^2 V_c^* (a^\dagger a + 1) V_c dx$$

$$= |c|^2 (|c|^2 + 1)$$

$$\Delta N = |c|$$

三. 无限深方势阱

$$V(x) = \begin{cases} \infty, & x < 0 \text{ 或 } x > 2a \\ 0, & 0 < x < 2a \end{cases}$$



$$t \text{ 时刻 } \psi(t) = A \sin\left(\frac{\pi x}{a}\right) \quad (0 < x < a)$$

(1) 求 A

归一化

$$\int_0^a A^2 \sin^2 \frac{\pi x}{a} dx = 1$$

$$A = \sqrt{\frac{2}{a}}$$

(2) 求 $\bar{E}$

$$\int \psi^* \hat{H} \psi dx = \frac{\pi^2}{2a^2}$$

$$\} \langle t | H | t \rangle = \langle 0 | H | 0 \rangle$$

所以算初始的就行

(3) 求可测出哪些能量

$$\text{本征态 } \psi_n = \sqrt{\frac{1}{a}} \sin \frac{n\pi x}{2a} \quad E_n = \frac{n^2 \pi^2}{8a^2}$$

$$C_n = \langle \psi_n | \psi \rangle = \frac{\sqrt{2}}{a} \int_0^a \sin \frac{n\pi x}{2a} \sin \frac{\pi x}{a} dx = \frac{4\sqrt{2} \sin \frac{n\pi}{2}}{(4-n^2)\pi}$$

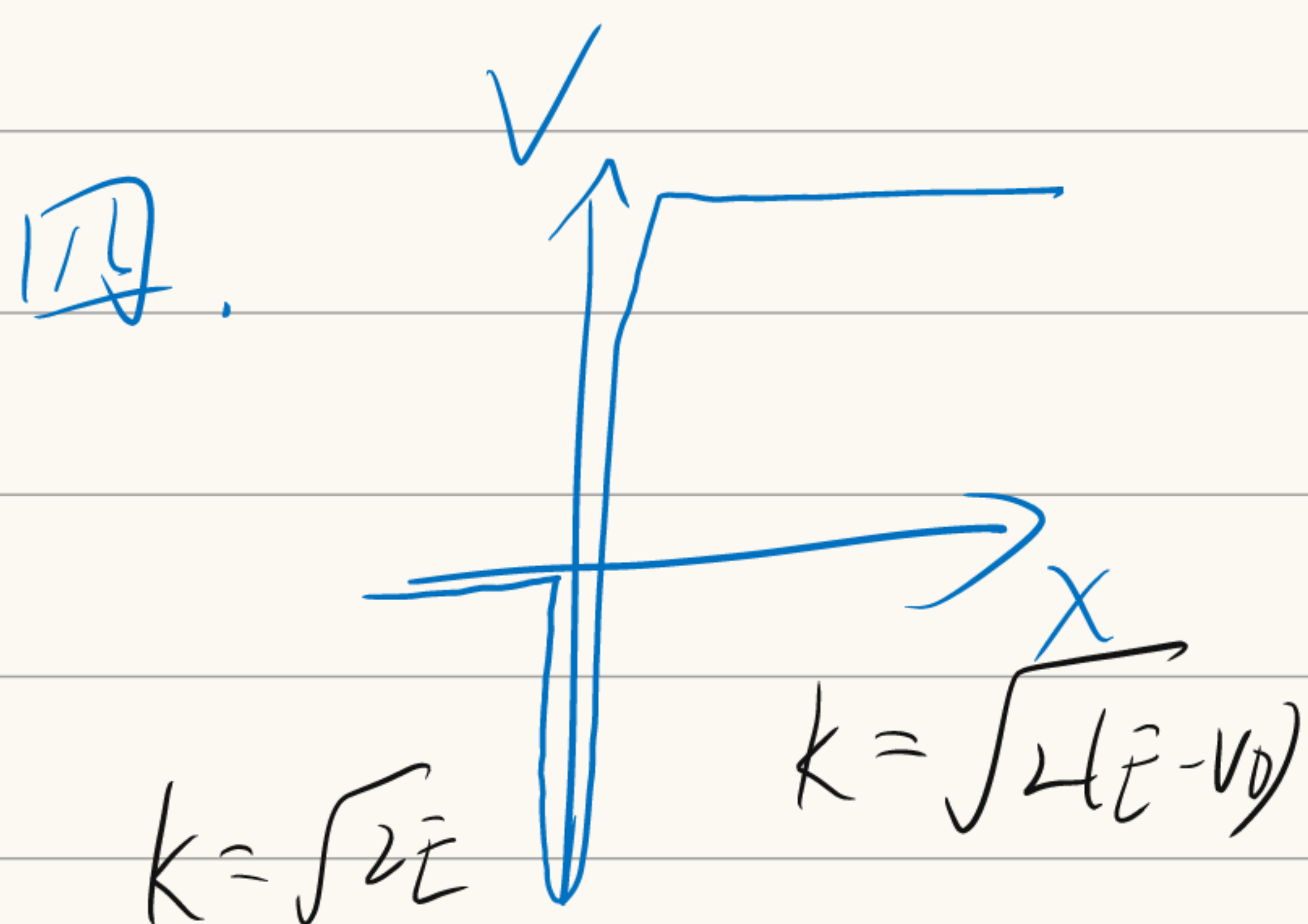
$$n=2 \quad C_2 = \frac{\sqrt{2}}{2}$$

$\therefore n$ 可取 $2, 2N+1, N \in \mathbb{N}^*$

(4) 求最大概率的能量和密度

(3')

$$n=2 \quad E = \frac{\pi^2}{2a^2} \quad \rho = \frac{1}{2}$$



$$V = \alpha \delta(x) + \begin{cases} 0, & x < 0 \\ V_0, & x > 0 \end{cases}$$

$$\psi(x) = \begin{cases} A e^{ikx} + B e^{-ikx} & 4/ \\ F e^{ikx} & 4/ \end{cases}$$

$$A + B = F \quad 4'$$

$$-\frac{1}{2} (\psi'|_{0+} - \psi'|_{0-}) + \alpha \psi(0) \quad 4'$$

结果

$$T = \frac{|F|^2}{|A|^2} \frac{k_2}{k_1} = \frac{4k_1 k_2}{(k_1 + k_2)^2 + 4\alpha^2} \quad 4'$$

